

Airline Review Sentiment Analysis

Lindy Bujak
Stats 507
University of Michigan
Ann Arbor, MI
lindy@umich.edu

Abstract—Sentiment analysis of customer reviews is an important natural language processing task because it converts unstructured text into actionable information about user experience and service quality. In this project, airline review text from the TCRG/airline_reviews dataset was classified into three sentiment categories: negative, neutral, and positive. After exploratory data analysis, three pretrained transformer models (DistilBERT, BERT, and RoBERTa) were fine-tuned for multi-class text classification using Hugging Face Transformers. Model performance was evaluated on held-out data using accuracy and macro F1 score. Among the models tested, RoBERTa achieved the strongest performance, reaching approximately 0.85 accuracy and 0.85 macro F1. The results show that transformer-based models are effective for short airline review text and that model architecture affects downstream classification performance. This project demonstrates a complete workflow for dataset exploration, transformer fine-tuning, and quantitative evaluation in Python.

Index Terms—sentiment analysis, natural language processing, transformers, BERT, RoBERTa, DistilBERT, text classification

I. INTRODUCTION

Sentiment analysis is a widely used natural language processing (NLP) task that aims to identify the emotional polarity of text, such as whether a review is negative, neutral, or positive. In business settings, sentiment analysis is especially valuable because customer feedback often contains information about product quality, delays, service interactions, and overall satisfaction. However, review text is unstructured and noisy, which makes it difficult to analyze using traditional structured-data methods alone.

Recent advances in NLP have been driven by transformer-based language models. Unlike earlier approaches such as bag-of-words representations and conventional machine learning classifiers, transformer models use contextualized representations that capture the meaning of words based on surrounding context. Models such as BERT, RoBERTa, and DistilBERT have therefore become strong baselines for many text classification tasks, including sentiment analysis.

The goal of this project is to compare the performance of several pretrained transformer models on airline review sentiment classification and determine which architecture is most effective for this dataset. Specifically, this work evaluates DistilBERT, BERT, and RoBERTa on a three-class classification task using short airline reviews. In addition to comparing predictive performance, the project also examines characteristics of the dataset through exploratory data analysis,

including class balance, review length, and common words associated with different sentiment labels.

Prior literature supports the use of transformer-based methods for sentiment analysis. Devlin et al. introduced BERT in their paper *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*, which established strong performance across a wide range of NLP tasks through bidirectional contextual pretraining [1]. Liu et al. later showed in *RoBERTa: A Robustly Optimized BERT Pretraining Approach* that optimized pretraining procedures can improve upon BERT [2]. Sanh et al. introduced DistilBERT in *DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter*. This describes a smaller and faster distilled variant that preserves much of BERT’s performance while reducing computational cost [3]. More recently, Jim et al.’s *Recent advancements and challenges of NLP-based sentiment analysis: A state-of-the-art review* surveyed recent developments in NLP-based sentiment analysis and emphasized the importance of pretrained transformer models and domain-specific fine-tuning for strong sentiment classification performance [4]. Motivated by this literature, this project applied and compared these transformer architectures on airline review data.

II. METHOD

A. Problem Formulation

This project considers a supervised multiclass text classification problem. The input x is the text of an airline review, and the output $y \in \{0, 1, 2\}$ is the sentiment label, where 0 denotes negative, 1 denotes neutral, and 2 denotes positive. The objective is to learn a mapping from review text to sentiment category that generalizes well to unseen examples.

The dataset used is TCRG/airline_reviews, obtained through the Hugging Face Datasets library. The reviews are short pieces of user-generated text labeled with one of the three sentiment classes. Before modeling, exploratory data analysis was performed to inspect class distribution, missing values, and review lengths. Word-count histograms and class-specific word clouds were also generated to better understand the structure and vocabulary of the dataset.

B. Preprocessing

Text preprocessing was performed using a Hugging Face tokenizer corresponding to each selected model checkpoint. For each experiment, the review text was tokenized with truncation enabled so that the text could be converted into

the input format expected by the transformer model. The tokenizer mapped text into token IDs and related attention inputs while preserving the pretrained vocabulary associated with the selected architecture.

C. Modeling

Three pretrained transformer models were evaluated: DistilBERT, BERT, and RoBERTa. Each model was loaded using `AutoModelForSequenceClassification` with the number of output labels set to three. Fine-tuning was then carried out on the airline review dataset so that the pretrained language representations could adapt to the sentiment classification task.

Training was implemented with the Hugging Face `Trainer` API. The training pipeline included model specification, training arguments, tokenized training and validation datasets, a dynamic padding data collator, and a custom metric function. Hyperparameters such as learning rate, batch size, number of epochs, and weight decay were defined through `TrainingArguments`. The same general training pipeline was repeated for each model so that their downstream performance could be compared consistently.

D. Evaluation

Model performance was evaluated using accuracy and macro F1 score. Accuracy measures the overall fraction of correctly classified reviews, while macro F1 gives equal weight to each class and is therefore appropriate for multiclass sentiment evaluation. After training, predictions were generated on held-out data, and additional diagnostic outputs such as classification reports and confusion matrices were used to interpret model behavior across the three sentiment classes.

This methodology provides a complete end-to-end workflow consisting of dataset exploration, transformer-based preprocessing, supervised fine-tuning, and quantitative model comparison for airline review sentiment classification.

III. RESULTS

The sentiment classification pipeline was implemented using the Hugging Face Transformers framework. The dataset was split into training, validation, and test sets containing 5,104, 1,276, and 709 reviews, respectively. DistilBERT, BERT, and RoBERTa were fine-tuned for the same three-class classification task. Each model was trained for three epochs and evaluated using validation loss, accuracy, and macro F1, followed by test-set classification reports and confusion matrices.

Across the three models, validation performance was strong and fairly close, but RoBERTa achieved the best overall results. DistilBERT reached a final validation accuracy of 0.8386 with a macro F1 of 0.8384, while BERT improved slightly to 0.8448 accuracy and 0.8447 macro F1. RoBERTa produced the highest validation performance, with 0.8519 accuracy and 0.8522 macro F1. RoBERTa also achieved the lowest validation loss among the final model checkpoints, suggesting the strongest overall fit to the sentiment classification task.

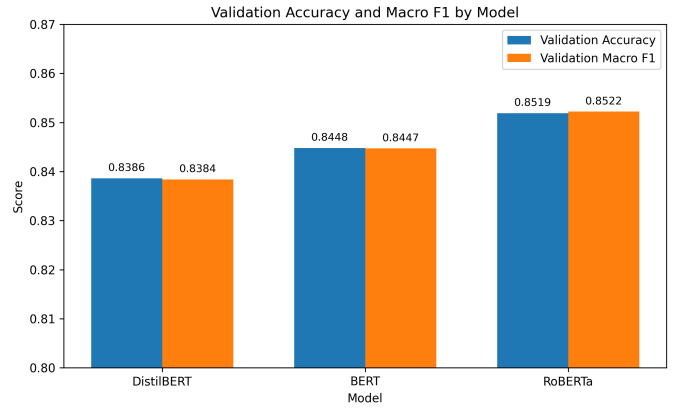


Fig. 1. Validation accuracy and macro F1 for DistilBERT, BERT, and RoBERTa on the airline review sentiment classification task.

As shown in Fig. 1, RoBERTa achieved the highest validation accuracy and macro F1 among the three models, although the performance gap between models was modest. DistilBERT completed training in about 3,692 seconds, compared with about 8,118 seconds for BERT and 7,341 seconds for RoBERTa. This means DistilBERT delivered competitive performance at substantially lower computational cost, making it an attractive choice when training efficiency is important. In contrast, RoBERTa offered the strongest predictive performance, which makes it the best model in this project when accuracy is prioritized over runtime.

Test-set results showed a similar pattern and revealed that the neutral class was the most difficult category for all three models. For DistilBERT, the neutral class had an F1 score of 0.75, compared with 0.83 for negative and 0.84 for positive. BERT also produced an F1 of 0.75 for the neutral class, while negative and positive classes reached 0.83 and 0.86, respectively. RoBERTa showed the best balance across classes, but neutral sentiment remained the weakest category with an F1 score of 0.78, compared with 0.85 for negative and 0.84 for positive. This suggests that neutral airline reviews overlap more with both positive and negative language, making them inherently harder to distinguish.

As shown in Fig. 2, many of the model’s errors involved neutral reviews being predicted as negative or positive rather than remaining in the neutral class. This pattern supports the earlier class-level results and helps explain why neutral sentiment was the most difficult category for all three models. RoBERTa reduced these neutral-class errors more effectively than DistilBERT and BERT, which helps explain its stronger macro F1 score. Overall, the results indicate that pretrained transformer models are highly effective for short airline-review sentiment classification, and that RoBERTa provided the best overall tradeoff between class balance and predictive accuracy in this project.

IV. CONCLUSION

This project developed and evaluated a transformer-based sentiment analysis pipeline for airline review classification.

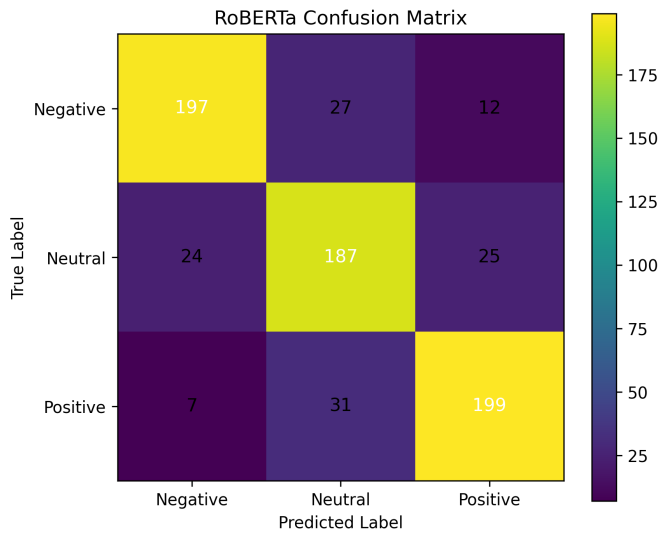


Fig. 2. Confusion matrix for the RoBERTa model on the test set. Most errors occur between the neutral class and the other two sentiment classes.

Using the TCRG/airline_reviews dataset, three pre-trained models (DistilBERT, BERT, and RoBERTa) were fine-tuned to classify reviews into negative, neutral, and positive sentiment categories. Exploratory data analysis, tokenizer-based preprocessing, supervised fine-tuning, and quantitative evaluation were all carried out within a consistent Hugging Face workflow.

The results showed that all three transformer models performed well on this task, with validation accuracy above 0.83 in every case. DistilBERT achieved a validation accuracy of 0.8386 and macro F1 of 0.8384, BERT achieved 0.8448 accuracy and 0.8447 macro F1, and RoBERTa produced the strongest overall performance with 0.8519 accuracy and 0.8522 macro F1. On the test set, RoBERTa also provided the best balance across classes, while the neutral class remained the most difficult for all models to classify. These results met the project objective of comparing transformer architectures and identifying the most effective model for airline review sentiment analysis.

One important limitation of this project is the nature of the dataset itself. The reviews are relatively short and the sentiment classes are evenly balanced, which makes the task cleaner than many real-world sentiment analysis problems. As a result, model performance on this dataset may overestimate how well the models would generalize to longer, noisier, or more imbalanced customer review data. In addition, only a small set of transformer architectures and training settings were compared, so further hyperparameter tuning or experiments with larger domain-specific models could improve performance.

Overall, this project demonstrates that pretrained transformer models are highly effective for short-text airline sentiment classification. Among the models tested, RoBERTa was the best-performing architecture, while DistilBERT offered

a strong efficiency-performance tradeoff. Future work could explore more extensive hyperparameter tuning, additional pre-trained models, or evaluation on more realistic airline review datasets to further improve robustness and generalization.

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